

Predicting Health-Service Needs and Outcomes

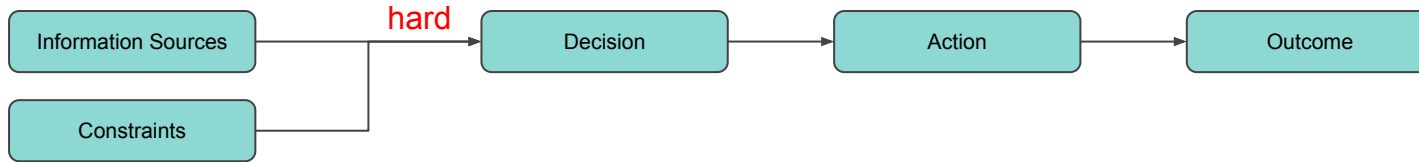
Fabian Egersdörfer, 08.10.26



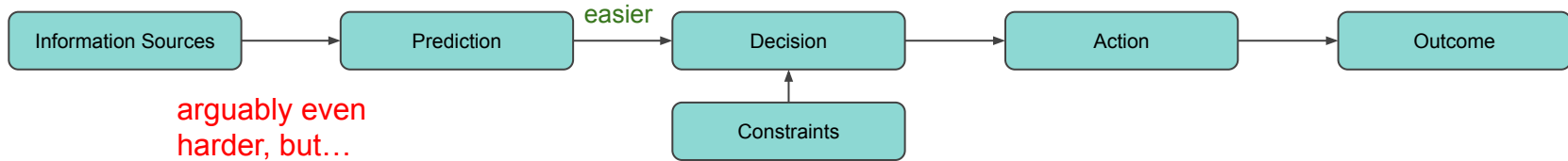


The Value of Prediction

Clinical and Policy decision making requires answering hypotheticals based on incomplete and hard to interpret information



Is there an **observable** quantity that would make the decision problem easier?

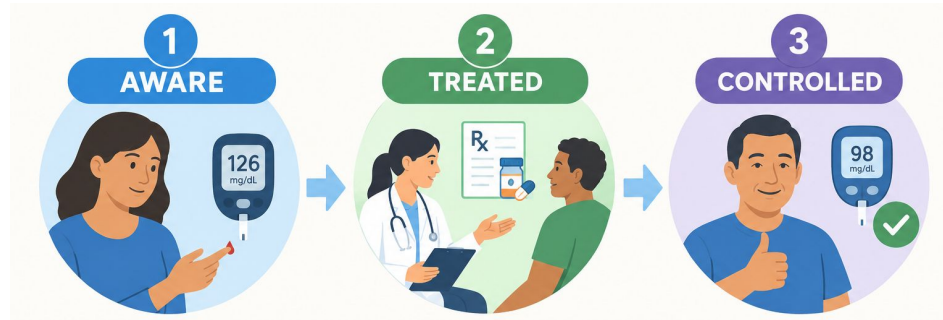


Prediction turns a decision problem into an **inference problem**

- + General purpose methodology
- + Objective performance metrics
- + Explainability
- + Access to information sources evading human intuition

- Data collection effort
- Generalizability is limited
- Access to information sources evading human intuition

A Running Example: Diabetes Care Cascade



Local health service availability

Existence and capacity of services to provide diabetes detection and care.

- Number and type of facilities (e.g., primary care clinics, hospitals, screening sites)
- Availability of diagnostic tests (e.g., blood glucose, HbA1c)
- Availability of trained health workforce
- Service hours and operating days



Local health service accessibility

Ease of reaching and using available services.

- Geographic distance to facilities
- Travel time and transport options
- Cost of services (out-of-pocket costs, insurance coverage)
- Waiting times for appointments
- Physical accessibility (e.g., for people with disabilities)
- Integration with other community services



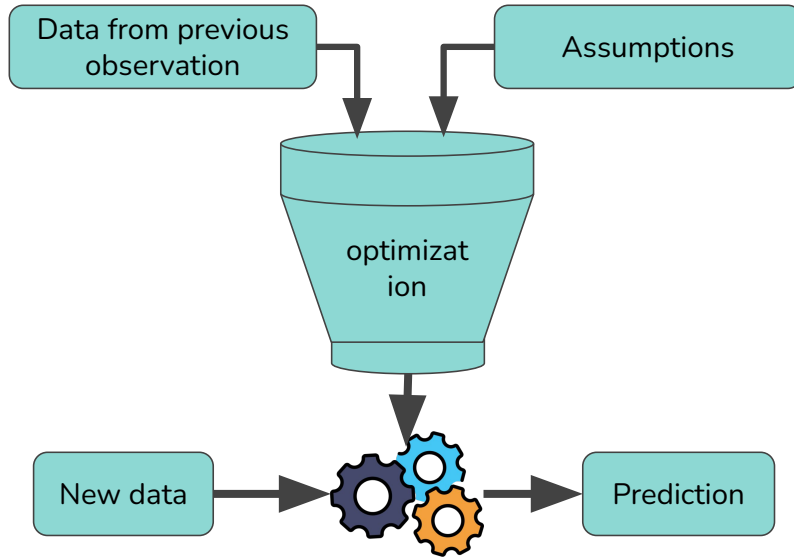
Local health service acceptability

Extent to which services are trusted, culturally appropriate, and aligned with community needs.

- Community trust in health services and providers
- Cultural and language appropriateness
- Perceived quality of care
- Stigma related to diabetes or using services
- Patient attitudes and preferences
- Previous experiences and satisfaction with care



Prediction Is Not Causation - Prediction is an Inference



Assumptions can be wrong as long as the model provides useful results!

Inferences have a clear set of **assumptions** (the model) and **remaining uncertainty** (the models parameters) and a measure for how suitable a set of parameters is for describing past **observations** (the **likelihood**)

The answer to an inference problem is a distribution not a number



Beware the causality trap: We rely on similarity not cause and effect

Lack of causality means you have to be careful about what question you are asking:



Who is likely to be readmitted ?
(Observable inference)



Who will benefit from additional follow-up ?
(Causal inference - Dangerous Assumption)

Example:

Older patients may have higher predicted risk. That does **not** necessarily mean that giving older patients more follow-up will reduce their risk.

Similarity is the mother of all assumptions. Assumptions can be wrong as long as the model provides useful results, but there is no such thing as causal inference.



Define the exact problem before choosing a model

1	Setting
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2	Potential Action
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3	Prediction Target
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4	Prediction Time
---	-----------------

5	Data
---	------

6	Model
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Our Example (Step III)

1	Setting Treatment planning after diagnosis
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2	Potential Action Additional follow up / behavioural intervention
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3	Prediction Target Step III (well adjusted medication)
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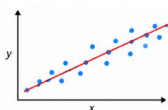
4	Prediction Time after diagnosis
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5	Data General info, diagnostic results from patients that made step II
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6	Model ???
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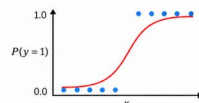
Inference in practice

Linear Regression



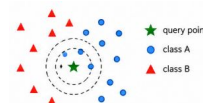
Continuous outputs

Logistic Regression



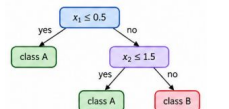
Binary outputs (probabilistic)

K Nearest Neighbor



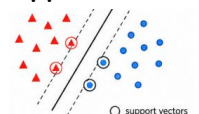
If you have good distance measure

Decision/Density Trees



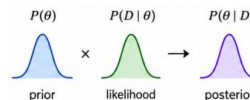
Usually the performance winner

Support Vector Machine



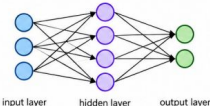
(more) robust generalization

Bayesian Inference



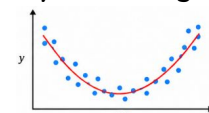
Modeling uncertainty

Neural Network



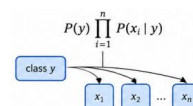
Is it worth the pain ?

Polynomial Regression



Only if you know why

Naïve Bayes



Do not dismiss

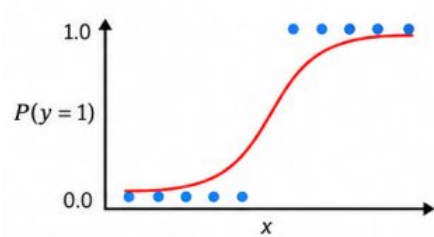
Use complex models for complex problems. Use simple models for simple problems.



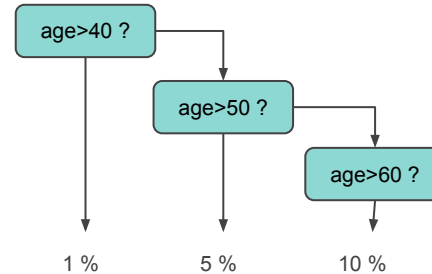
What is complexity ? - Example A

Predicting stroke risk from patient age:

Model 1 - Logistic Regression



Model 2 - Tree Model



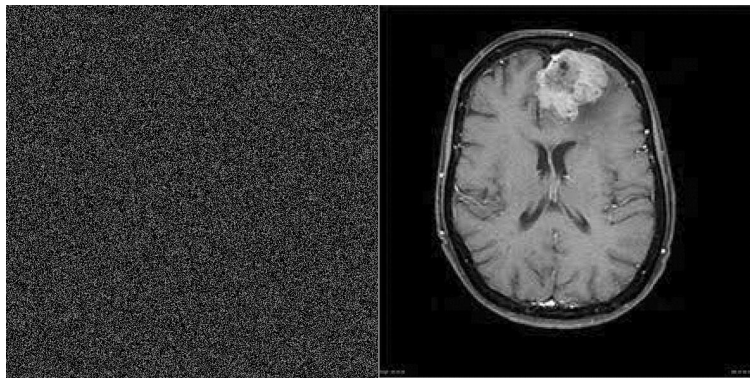
What if the model is presented with a patient at age 10 ?

In what scenarios would the tree model have the advantage ?



What is complexity ? - Example B

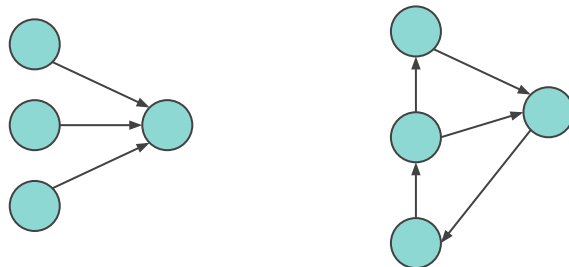
Predicting tumor growth from MRI:



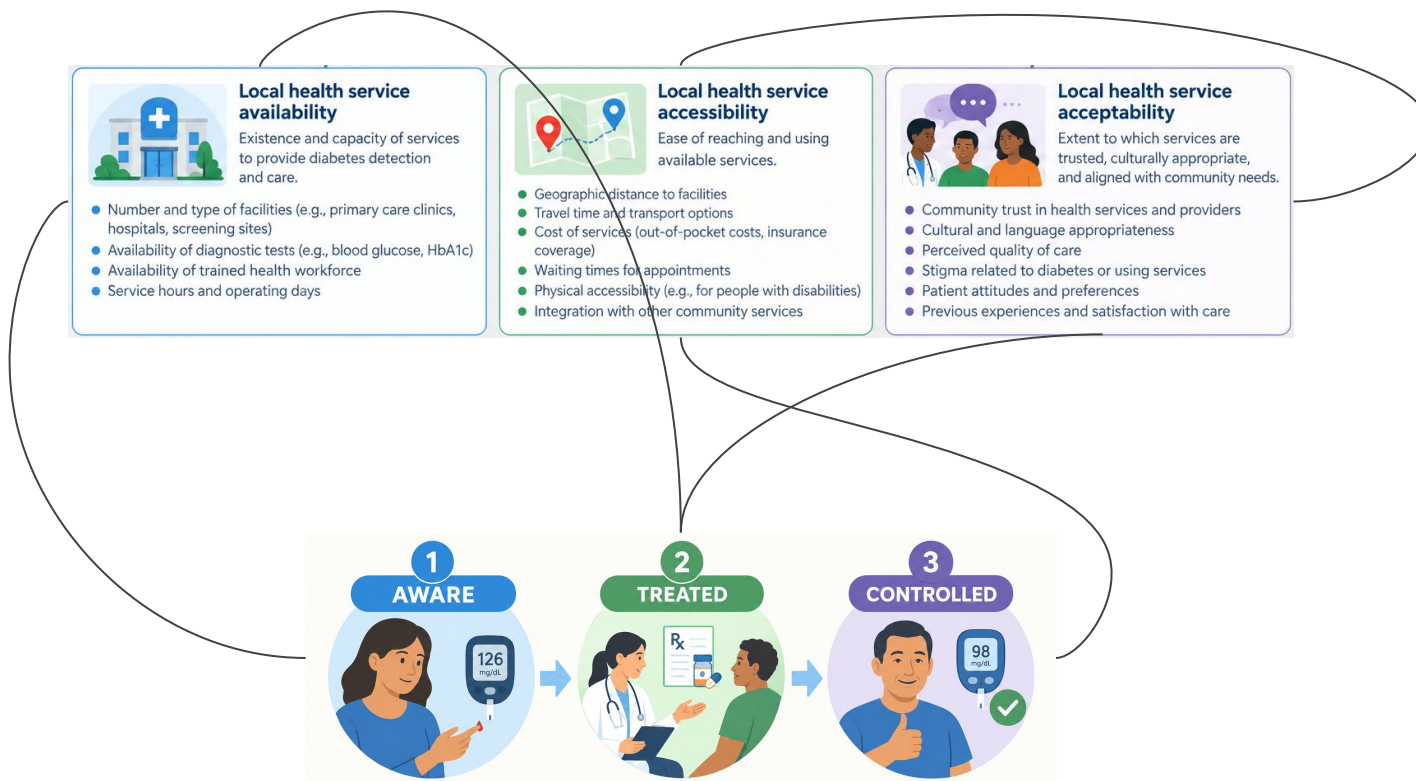
Same data different meaning !

Complexity:

- **Interdependence** rather than size
- Contextual understanding required

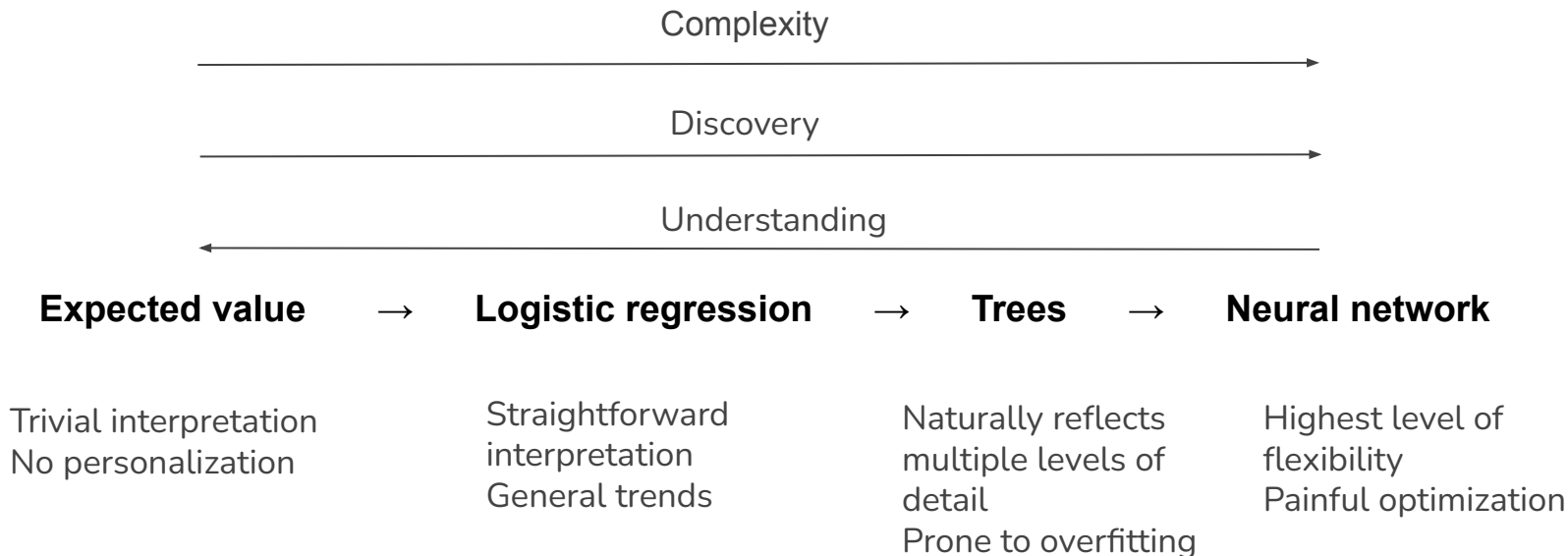


Our example: Complex or not?





Introduce complexity in Layers



More complex models should **refine not improve**



Important Evaluation Metrics (for binary classification)

Question	Example measure	Interpretation
Can we rank high- and low-risk patients?	AUROC	0.5 : model is guessing 1.0 : pos./neg. cases can be separated
Are predicted probabilities accurate, i.e. is it consistency with the true frequency among similar patients?	Calibration	Model produces sensible results on population level
Does the model identify useful patients at a threshold?	Sensitivity	
Does using it improve decisions under certain priorities?	Decision/utility analysis Net benefit	
Does it work elsewhere?	External validation	

Predicting Roulette role:
AUROC = 0.501 means you're rich

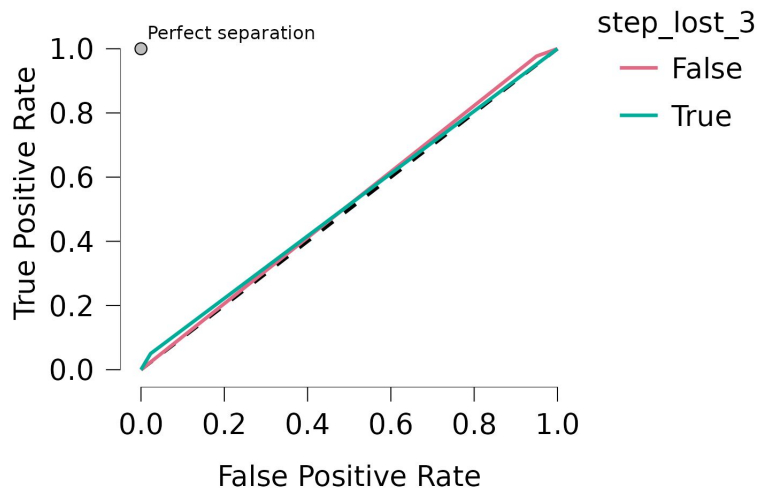
evaluation depends on the context

Reading out computer memory:
AUROC = 0.998 means you have
a useless piece of trash



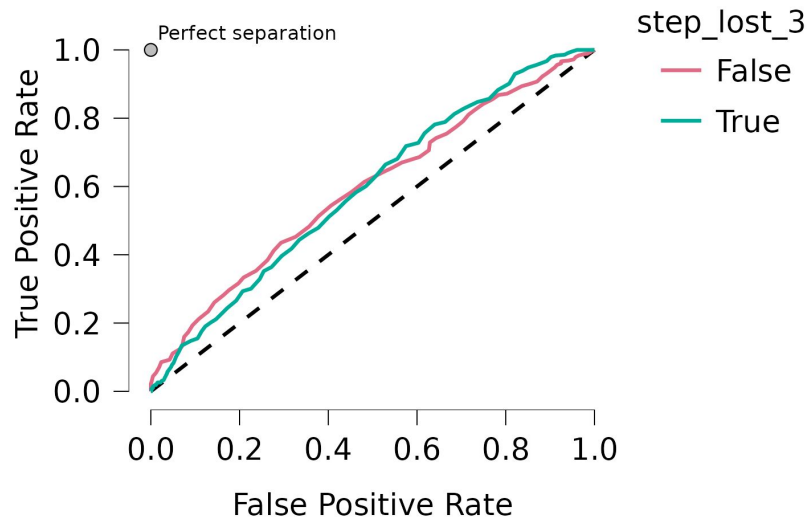
Our example

Logistic Regression



AUROC = 0.5

Decision Tree



AUROC = 0.6



How models fail

Treating variance as information (overfitting)

Prevention Strategy:
Internal validation
(Cross validation / Hold-out
validation)

Countermeasure:
Regularization

Violated assumptions (Generalization failure)

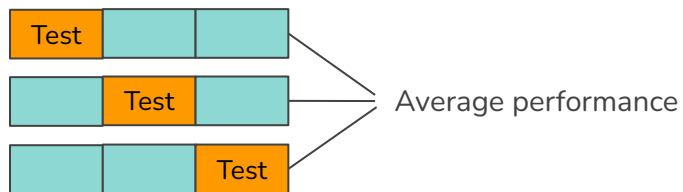
Prevention Strategy:
External validation
(Geographic and temporal)

Countermeasure:
e.g. Recalibration

Generalization is an empirical question and is not guaranteed by high test score.

Internal validation

Cross-Validation



Hold-out validation



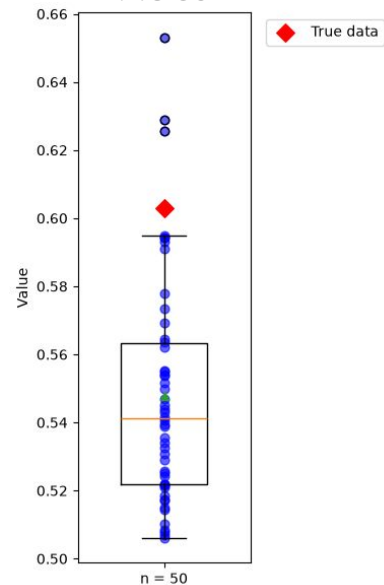
Resampling techniques



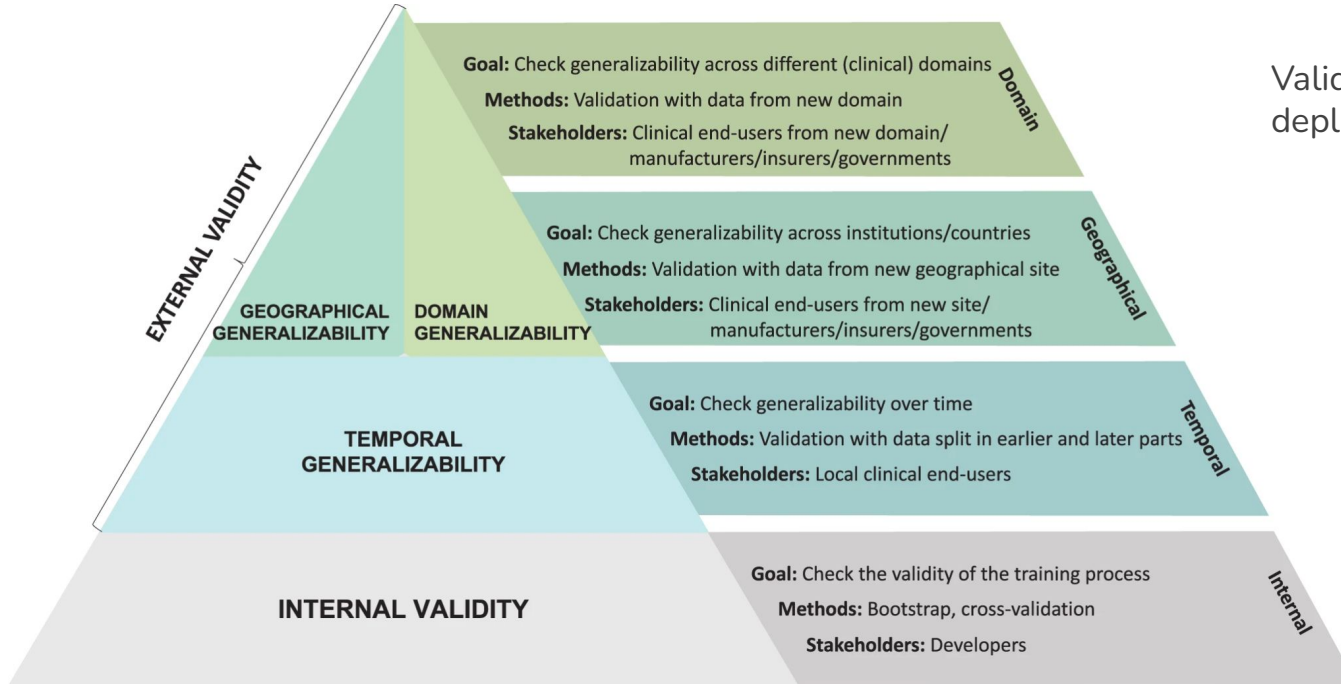
shuffle
data



AUCs

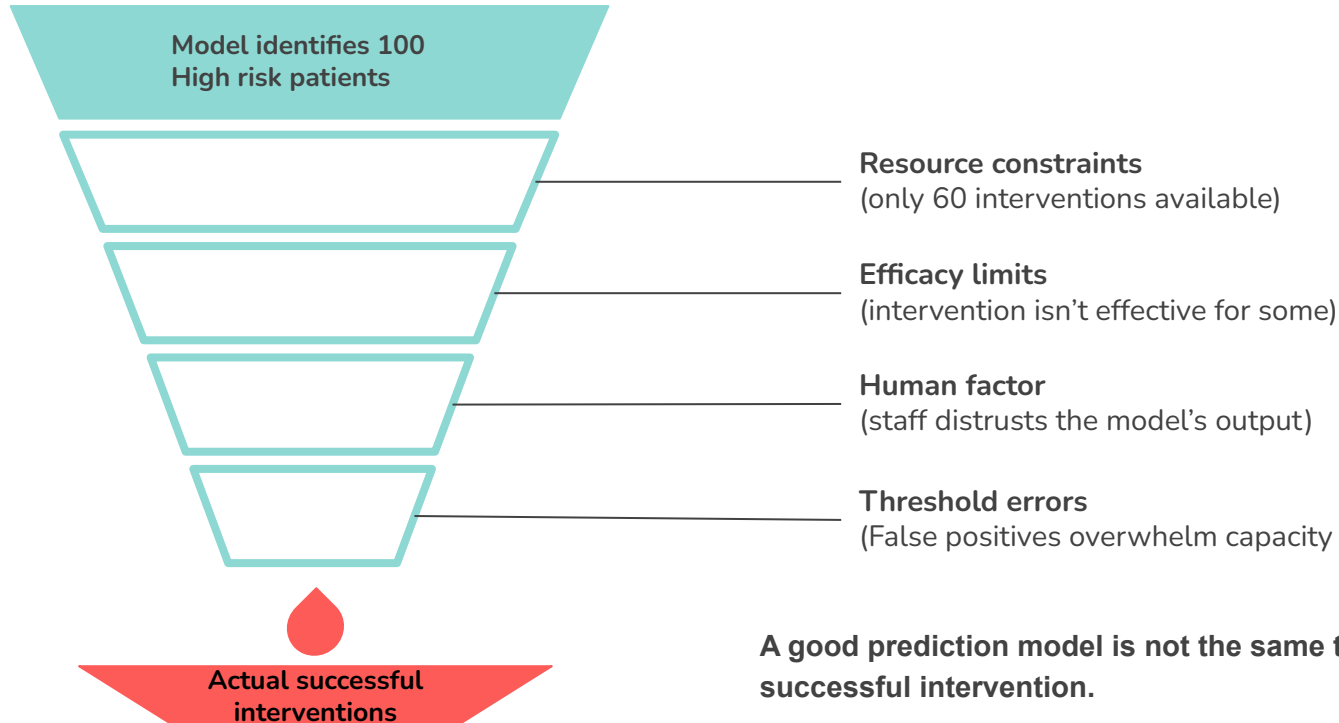


External Validation



Validate as if you are
deploying !

And now what? From Prediction to Action





Take Home Messages

1. Define the prediction problem by its associated decision problem

Who? When? What? Over what horizon? For what decision?

2. Always compare against a sensible baseline

Complexity has to earn its place. Simplicity is a value.

3. Validate according to the deployment scenario

Random train/test splits aren't always enough.

4. Prediction is not intervention

Identifying high-risk patients doesn't guarantee that intervening will help.

5. Predictions are Inferences

Thanks for your attention!

Fabian Egersdörfer

